Topoi (ILF)

Goldblatt / Exercises / Reading group

Summer 2026

Contents

Part I

Introduction

Meeting II

Category Theory Pt. 1

1.1 Minimal Exercises

1.1.1 Pre-order Category

For each natural number n, consider the pre-order category $\mathbf{n}$ associated with the pre-order $0 \leq 1 \leq \ldots \leq n-1$. How many arrows does $\mathbf{n}$ have?

1.1.2 Subcategories of a Monoid

Consider the monoid $\mathbf{N} = (N, +, 0)$ of natural numbers, which gives rise to a category whose collection of objects is the singleton $\{N\}$ and whose collection of arrows is the set N of natural numbers. Which of the following $\mathscr{C}$ are valid subcategories?

- The empty category $\mathscr{C} = \mathbf{0}$
- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{0\} = \{1, 2, 3, 4, ...\}$
- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{1\} = \{0, 2, 3, 4, ...\}$
- $\mathscr{C}$-objects are $\{N\}$, and $\mathscr{C}$-arrows are $N \setminus \{2\} = \{0, 1, 3, 4, ...\}$

1.2 Exercises for Good Preparation

1.2.1 Endomorphism Monoid

Let $\mathscr{C}$ be a category and X be an object of $\mathscr{C}$. Verify that the collection $\mathscr{C}(X, X)$ of all $\mathscr{C}$-arrows from X to X forms a monoid with respect to composition.

1.2.2 Sum Category

Let $\mathscr{C}$ and $\mathscr{D}$ be categories. The sum category, denoted $\mathscr{C} + \mathscr{D}$, is a category whose collection of objects is the disjoint union of $\mathscr{C}$-objects and $\mathscr{D}$-objects, and whose collection of arrows is the disjoint union of $\mathscr{C}$-arrows and $\mathscr{D}$-arrows.

- Describe how to inherit domains, codomains, identity, and composition from both $\mathscr{C}$ and $\mathscr{D}$. Verify that $\mathscr{C} + \mathscr{D}$ is a category.
- Verify that there are no $(\mathscr{C} + \mathscr{D})$-arrows between any object that came from $\mathscr{C}$ and any object

that came from $\mathscr{D}$.

- Find an example of $\mathscr{C}$ and $\mathscr{D}$ where if set union was used instead of disjoint union, then the resulting construction would not yield a category.

1.3 Extra Exercise

1.3.1 Quivers

A *quiver* is a directed graph that allows loops and multiple parallel edges. In particular, a quiver Q comprises:

- a collection of things called Q-*vertices*;
- a collection of things called Q-*edges*;
- operations assigning to each Q-edge f a Q-vertex $s(f)$ (the "source" of f) and a Q-vertex $t(f)$ (the "target" of f).

Every category has an underlying quiver by forgetting the composition (and identity): each object is a vertex, each arrow is an edge, $s(f) = \mathrm{dom}(f)$, and $t(f) = \mathrm{cod}(f)$.

Two quivers are considered isomorphic when there is a bijection of vertices and a bijection of edges that preserves sources and targets. Two categories are considered isomorphic when there exists an isomorphism of the underlying quivers that additionally preserves composition and identity.

Find an example of two non-isomorphic categories such that the underlying quivers are isomorphic.

Meeting III

Category Theory Pt. 2

2.1 Minimal Exercises

2.1.1 Iso Arrows

Exercises from Goldblatt section 3.3: in any category, verify the following:

1. Every identity arrow is iso.
2. If $b \xrightarrow{f} c$ is iso, so is $c \xrightarrow{f^{-1}} b$.
3. $a \xrightarrow{f \circ g} c$ is iso if $b \xrightarrow{f} c$, $a \xrightarrow{g} b$ are, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

2.1.2 Terminal Objects

Selected exercises from Goldblatt section 3.6:

1. For any category $\mathscr{C}$, prove that all terminal $\mathscr{C}$-objects (if any) are isomorphic (to each other).
2. Find terminals in $\mathbf{Set}^2$, $\mathbf{Set}^{\rightarrow}$, and the poset $\mathbf{n}$.

2.1.3 Products

Selected exercises from Goldblatt section 3.8:

Let a, b be two $\mathscr{C}$-objects that have a product $a \times b$ with projections $\mathrm{pr}_a : a \times b \to a$ and $\mathrm{pr}_b : a \times b \to b$.

Exercise 1: Prove that $\langle \mathrm{pr}_a, \mathrm{pr}_b \rangle = 1_{a \times b}$.

Exercise 2: Prove that if $\langle f, g \rangle = \langle k, h \rangle$, then $f = k$ and $g = h$.

Exercise 3: Prove that $\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h$.

Exercise 4: Show that if $\mathscr{C}$ has a terminal object and products, then for any $\mathscr{C}$-object a, $a \cong a \times 1$ and indeed $\langle 1_a, !_a \rangle$ is iso.

Exercise 5: Prove that $1_a \times 1_b = 1_{a \times b}$.

Exercise 6: Prove that a product $b \times a$ exists and that $a \times b \cong b \times a$.

2.2 Exercises for Good Preparation

2.2.1 Monic Arrows

Exercises from Goldblatt section 3.1: in any category, consider arrows $a \xrightarrow{f} b, b \xrightarrow{g} c$ as in the commutative diagram:

$$\begin{tikzcd} a \arrow[r, "f"] \arrow[dr, "g\circ f"'] & b \arrow[d, "g"] \\ & c \end{tikzcd}$$

(1) Prove that if f and g are both monic, then $g \circ f$ is monic.

(2) Prove that if $g \circ f$ is monic, then f is monic.

2.2.2 Terminal Objects

Remaining exercise from Goldblatt section 3.6:

3. Show that an arrow $1 \to a$ whose domain is a terminal object must be monic.

2.2.3 Isomorphic Objects

Exercise 1 from Goldblatt section 3.4: For any $\mathscr{C}$-objects a, b, c in any category $\mathscr{C}$, prove the following:

(i) $a \cong a$

(ii) if $a \cong b$, then $b \cong a$

(iii) if $a \cong b$ and $b \cong c$, then $a \cong c$

2.2.4 Products

More exercises from Goldblatt section 3.8:

Exercise 7: Show that $(a \times b) \times c \cong a \times (b \times c)$.

Exercise 8(i): Show that $(f \times h) \circ \langle g, k\rangle = \langle f \circ g, h \circ k\rangle$.

Exercise 8(ii): Show that $(f \times h) \circ (g \times k) = (f \circ g) \times (h \circ k)$.

2.3 Extra Exercises

2.3.1 Isomorphic Objects

Exercise 2 from Goldblatt section 3.4: Prove that $\mathbf{Finord}$ is a skeletal category.

($\mathbf{Finord}$ is the full subcategory of $\mathbf{Set}$ with finite ordinals as objects; refer to Goldblatt section 2.5 example 9 for further details if necessary.)

2.3.2 Products

Remaining exercises from Goldblatt section 3.8:

Exercise 9: Analyze in detail the formation of the projection arrows $\mathrm{pr}_1^m, \ldots, \mathrm{pr}_m^m$, and verify assertions relating to the last diagram (refer to Goldblatt). Show that for any product arrow

$$c \xrightarrow{\langle f_1,\ldots,f_m\rangle} a^m,$$

we have $\mathrm{pr}_j^m \circ \langle f_1, \ldots, f_m\rangle = f_j$, for $1 \leq j \leq m$.

Exercise 10: Develop the notion of the product $a_1 \times a_2 \times \cdots \times a_m$ of m objects (possibly different) and the product of m arrows (possibly different).

2.3.3 Co-products

Exercises from Goldblatt section 3.9:

Exercise 1: Show that $A + B, i_A, i_B$ as just defined (in the category $\mathbf{Set}$, refer to Goldblatt) satisfy the co-product definition.

Exercise 2: If $A \cap B = \emptyset$, show $A \cup B \cong A + B$.

Exercise 3: Define the co-product arrow $f + g : a + b \to c + d$ of arrows $f : a \to c$ and $g : b \to d$ and dualise all of the Exercises in section 3.8 (see Goldblatt).

(Write out the dual statement of each exercise; you do not need to write out a dual proof. Though it can feel tedious, this exercise helps develop intuition about dual concepts. It can be insightful to see how a concept and its dual may be fundamentally different constructions, yet behave categorically similar.)

2.3.4 Jointly Monic Pair

Let a, b be $\mathscr{C}$-objects. Suppose that we have a "span from a to b", which is a diagram in $\mathscr{C}$ of the following shape:

$$\begin{tikzcd} & r \arrow[dl, "f"'] \arrow[dr, "g"] & \\ a & & b \end{tikzcd}$$

We say that $\{f, g\}$ is **jointly monic** if for all parallel pairs $h, k : x \to r$, the following holds:

whenever $f \circ h = f \circ k$ and $g \circ h = g \circ k,$ then $h = k$

This is a generalization of being "left-cancellable" in the sense that both conditions need to hold true before we can simultaneously cancel out f and g.

Exercise: Suppose that the product $a \times b$ exists. Prove that $\langle f, g \rangle : r \to a \times b$ is monic if and only if $\{f, g\}$ is jointly monic.

Meeting IV

Category Theory Pt. 3

3.1 Minimal Exercises

3.1.1 Equalizers

Exercises from Goldblatt section 3.10:

Let $E \overset{f}{\rightarrowtail} A$ be a monomorphism (a.k.a. monic arrow) in **Set**. Define $h, g : A \to \{0, 1\}$ as follows:

$$h(x) = 1$$
$$g(x) = \begin{cases} 1 & \text{if } x \in \operatorname{Im} f \\ 0 & \text{if } x \notin \operatorname{Im} f \end{cases}$$

Exercise 1: Prove that f is the equalizer of g and h.

Exercise 2: Show that in a poset, the only equalizers are the identity arrows.

3.1.2 Pullbacks

Exercise from Goldblatt section 3.13: Show that if

$$\begin{array}{ccc} a & \xrightarrow{g} & b \\ \downarrow & & \downarrow \\ c & \xrightarrow{f} & d \end{array}$$

is a pullback square, and f is monic, then g is also monic.

3.1.3 Subobjects

Exercises from Goldblatt section 4.1:

Exercise 1: In **Set**, verify that $\mathrm{Sub}(D) \cong \mathscr{P}(D)$.

Exercise 2: Prove that $ev \circ \langle \ulcorner f \urcorner, x \rangle = f \circ x$ (Refer to Goldblatt for notation and context.)

If you have not read section 3.16 on exponentials, you may do exercise 2 just for **Set**.

3.1.4 Classifying subobjects

Exercises from Goldblatt section 4.2:

Exercise 1: Show that the character of $true : 1 \rightarrowtail \Omega$ is $\mathsf{1}_\Omega$

$$\begin{tikzcd}
1 \arrow[r, tail, "true"] \arrow[d] & \Omega \arrow[d, "\mathsf{1}_\Omega"] \\
1 \arrow[r, "true"] & \Omega
\end{tikzcd}$$

i.e. $\chi_{true} = \mathsf{1}_\Omega$.

Exercise 2: Show that $\chi_{\mathsf{1}_\Omega} = true_\Omega = true \circ \mathsf{l}_\Omega$.

$$\begin{tikzcd}
\Omega \arrow[r, tail, "\mathsf{1}_\Omega"] \arrow[d, "\mathsf{l}_\Omega"'] & \Omega \arrow[d, "true \circ \mathsf{l}_\Omega"] \\
1 \arrow[r, "true"] & \Omega
\end{tikzcd}$$

Exercise 3: Show that for any $f : a \to b$,

$$\begin{tikzcd}
a \arrow[rr, "f"] \arrow[dr, "true_a"'] & & b \arrow[dl, "true_b"] \\
& \Omega &
\end{tikzcd}$$

$true_b \circ f = true_a$.

3.2 Exercises for Good Preparation

3.2.1 Pushouts

Exercise from Goldblatt section 3.14: Dualise section 3.13.

(In particular, write out the concept of a pushout and write out the duals of examples 5 through 9, especially example 8 regarding the dual of the pullback lemma.)

3.2.2 Completeness

Exercises from Goldblatt section 3.15:

(1) Verify (B), and consider the details of that construction in **Set**. From Goldblatt, (B) states: given pullbacks and products, from a parallel pair $f, g : a \rightrightarrows b$ and a pullback of the form

$$\begin{tikzcd} d \arrow[r, "q"] \arrow[d, "p"'] & a \arrow[d, "\langle 1_a, g\rangle"] \\ a \arrow[r, "\langle 1_a, f\rangle"] & a \times b, \end{tikzcd}$$

it follows readily (from section 3.8) that $p = q$, and that this arrow is an equaliser of f and g.

(2) Show how to construct pullbacks from products and equalisers. (Refer to Goldblatt for a hint.)

(3) Dualise the Theorem of this section. (Refer to Goldblatt. You only need to write out the dual statement of the theorem.)

3.2.3 Exponentiation

Exercise from Goldblatt section 3.16: Prove part (5) of the Theorem, and interpret (1)–(5) as they apply to **Set**. For reference, the theorem is:

> **Theorem**
> Let $\mathscr{C}$ be a Cartesian closed category with an initial object 0. Then in $\mathscr{C}$,
>
> (1) $0 \cong 0 \times a$, for any object a;
>
> (2) if there exists an arrow $a \to 0$, then $a \cong 0$;
>
> (3) if $0 \cong 1$, then the category $\mathscr{C}$ is degenerate, i.e. all $\mathscr{C}$-objects are isomorphic;
>
> (4) any arrow $0 \to a$ with dom 0 is monic;
>
> (5) $a^1 \cong a, a^0 \cong 1, 1^a \cong 1$.

3.2.4 Topos

Exercise 1: Prove that in a topos:

- $\mathrm{Sub}(0) \cong 1$
- $\mathrm{Sub}(A + B) \cong \mathrm{Sub}(A) \times \mathrm{Sub}(B)$

If you need a hint for the second one, message us on Discord!

Exercise 2: in the topos $\mathbf{Set}^{\rightarrow}$, find the characteristic arrow of the following monic (from the left arrow to the right arrow):

$$\begin{tikzcd} \{0\} \arrow[r, hook] \arrow[d, hook] & \{0,1,2\} \arrow[d, hook] \\ \{0,1\} \arrow[r, hook] & \{0,1,2,3\} \end{tikzcd}$$

3.3 Extra Exercises

3.3.1 Pullbacks

Suppose

$$\begin{tikzcd} d \arrow[r, "q"] \arrow[d, "p"'] & b \arrow[d, "g"] \\ a \arrow[r, "f"] & c \end{tikzcd}$$

is a commutative diagram.

Exercise 1: Suppose the above is a pullback square. Prove that $\{p, q\}$ is jointly monic (refer to exercise 2.3.4 above for the definition).

Exercise 2: Suppose that p and g are both monic. Prove that the above square is a pullback iff for all commutative diagrams of the form

$$\begin{tikzcd} x \arrow[drr, bend left, "t"] \arrow[ddr, bend right, "s"'] & & \\ & & b \arrow[d, "g"] \\ & a \arrow[r, "f"] & c \end{tikzcd},$$

there exists an arrow $x \to d$ making the following diagram commute:

$$\begin{tikzcd} x \arrow[dr] \arrow[ddr, bend right, "s"'] & \\ & d \arrow[d, "p"'] \\ & a \end{tikzcd}$$

Exercise 3: Let $\mathscr{C}$ be a category with pullbacks. Prove that $\mathscr{C}$ has limits of the following shape:

$$\begin{tikzcd} \bullet \arrow[dr] & & \bullet \arrow[dl] \arrow[dr] & & \bullet \arrow[dl] \\ & \bullet & & \bullet & \end{tikzcd}$$

Exercise 4: Let $\mathscr{C}$ be a category in which the products $A \times C$ and $B \times D$ exist.

Suppose that the following diagram is a pullback:

$$\begin{tikzcd} R \arrow[r, "e"] \arrow[d, "\langle a,b \rangle"'] & S \arrow[d, "\langle c,d \rangle"] \\ A \times B \arrow[r, "f \times g"] & C \times D \end{tikzcd}$$

Prove that the following diagram is a limiting cone (for a diagram of shape given by the previous exercise):

$$
\begin{tikzcd}
A \arrow[d, "f"'] & R \arrow[l, "a"'] \arrow[r, "b"] \arrow[d, "e"] & B \arrow[d, "g"] \\
C & S \arrow[l, "c"] \arrow[r, "d"'] & D
\end{tikzcd}
$$

Does the converse hold?

Meeting V

Topos Structure Pt. 1

4.1 Minimal Exercises

4.1.1 Unique Morphism and Relation

Show that the unique morphism $\Omega^a \to \Omega^a$ corresponding to the relation $\in_a \rightarrowtail \Omega^a \times a$ is id_{Ω^a}.

4.1.2 Properties of Epimorphisms

From section 5.2:

- $f : a \to b$ is epic $\Leftrightarrow \exists g : f(a) \cong b$ such that $g \circ f^* = f$.

4.2 Exercises for Good Preparation

4.2.1 Left Ideals

Refer to section 4.6.

- Describe all the left-ideals in $(N, +, 0)$
- Show M is a group $\Leftrightarrow$ M and $\emptyset$ are the only left-ideals of $M(L_M = \{M, \emptyset\})$.

4.2.2 Some Structural Properties of Topoi

Refer to sections 5.1 - 5.2. Prove the following properties:

- *true* $: 1 \to \Omega$ equalizes $\mathrm{id}_\Omega : \Omega \to \Omega$ and *true*$_\Omega : \Omega \to \Omega$.
- If $p = q$, then f is epic.

4.3 Extra Exercise(s)

4.3.1 Power Objects in Topoi

Describe the power objects of the following Topoi:

- **Set**
- **Finset**
- **Finord** (the category of finite ordinals)
- **Set**^{2} (the category of pairs of sets)
- **Set**$^{\rightarrow}$ (the category of functions)
- **M-Set** (for some monoid M, an M-Set is the pair (X, λ) such that $\lambda : M \times X \rightarrow X$ is an action of M on X)
- **Set**$^{\mathcal{C}^{\text{op}}}$ (the category of presheaves)

Hint: The structure of the first five examples are given in detail in section 4.4. For the fifth example, read section 4.6 and this. For the final example, read this.